

The background is a solid blue color with a subtle grid pattern. Overlaid on this are faint, light-blue icons representing various data and database concepts: a line graph with three peaks, a pie chart with one slice highlighted, two circular progress indicators with percentage signs, and a document icon with a list of items. In the bottom right corner, there is a dark blue triangular shape.

DATABASE FUNDAMENTALS

ASSIGNMENT 2

ROUTE

PROBLEM 1

DreamHome Company is designed to connect individuals interested in renting properties with their owners. It's important to note that the term 'Rent' refers to the value assigned to the property, determined through negotiation between the customer and the owner.

To **achieve the (3NF)** using the provided data begin by detailing the steps for your answer. The primary key should be the Customer number. Exclude the Page number and Date from the bill shown below.

Page 1		DreamHome Customer Rental Details			Date 7-Oct-98	
Customer Name John Kay				Customer Number CR76		
Property Number	Property Address	Rent Start	Rent Finish	Rent	Owner Number	Owner Name
PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw

Customer_Rental Table								
Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-June-94	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

PROBLEM 2

The provided report contains detailed information concerning the organization's projects and the employees assigned to them. The records illustrate a many-to-many relationship: a project can involve multiple employees, and an employee can work on multiple projects. Additionally, each job classification is associated with a specific hourly charge rate (CHG/Hour). **Your task is to apply the First, Second, and Third Normal Forms (1NF, 2NF, and 3NF) to this data.**

PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS.	CHG/HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$ 85.50	23.8	\$ 2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson*	Database Designer	\$105.00	35.7	\$ 3,748.50
		106	William Smithfield	Programmer	\$ 35.75	12.6	\$ 450.45
		102	David H. Senior	Systems Analyst	\$ 96.75	23.8	\$ 2,302.65
				Subtotal			\$10,549.70
18	Amher Wave	114	Annelise Jones	Applications Designer	\$ 48.10	25.6	\$ 1,183.26
		118	James J. Frommer	General Support	\$ 18.36	45.3	\$ 831.71
		104	Anne K. Ramoras*	Systems Analyst	\$ 96.75	32.4	\$ 3,135.70
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	45.0	\$ 2,021.00
				Subtotal			\$ 7,172.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	65.7	\$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	\$ 96.75	48.4	\$ 4,682.70
		113	Delbert K. Joenbrood*	Applications Designer	\$ 48.10	23.6	\$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	\$ 26.87	22.0	\$ 591.14
		106	William Smithfield	Programmer	\$ 35.75	12.8	\$ 457.60
				Subtotal			\$13,660.10
25	Starlight	107	Maria D. Munzo	Programmer	\$ 35.75	25.6	\$ 879.45
		115	Travis B. Bawangi	Systems Analyst	\$ 96.75	45.8	\$ 4,431.15
		101	John G. News*	Database Designer	\$105.00	56.3	\$ 5,911.50
		114	Annelise Jones	Applications Designer	\$ 48.10	33.1	\$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
		118	James J. Frommer	General Support	\$ 18.36	30.5	\$ 559.98
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
				Subtotal			\$17,559.02
				Total			\$48,942.09

Note: * indicates project leader.